

BORTNAN, SwedenWMO: **023190**Lat: **62.7548N**Lon: **13.8427E**Elev: **1532**StdP: **13.90**Time Zone: **1.00 (EUC)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-15.3	-27.5	1.3	-19.2	-21.2	1.8	-14.9	15.3	33.0	13.8	31.6	1.5	290	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	75.8	59.1	72.5	58.6	68.7	56.8	62.1	70.9	60.2	68.1	58.2	65.4	5.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.8	78.4	64.4	56.8	72.7	62.7	54.9	67.8	61.2	28.6	70.9	27.2	68.0	25.8	65.2	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.9	10.4	-27.5	79.9	4.6	3.6	-30.8	82.4	-33.5	84.5	-36.1	86.6	-39.4	89.2		
(5)			-27.8	64.0	4.5	2.2	-31.0	65.6	-33.7	66.9	-36.2	68.1	-39.5	69.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.3	14.9	17.1	23.8	33.7	42.7	50.1	55.1	51.9	44.7	34.5	24.6	17.4
	DBStd	16.57	13.42	12.25	10.14	6.53	6.69	5.79	5.43	5.18	5.68	7.12	9.45	13.46
	HDD50	6109	1087	921	813	489	246	70	14	40	174	482	762	1011
	HDD65	11206	1552	1341	1278	939	692	446	308	407	610	945	1212	1476
	CDD50	379	0	0	0	0	18	73	174	98	14	2	0	0
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH74	170	0	0	0	0	9	42	102	17	0	0	0	0
	CDH80	10	0	0	0	0	0	1	9	0	0	0	0	0
Wind	WSAvg	4.1	3.6	3.8	4.6	4.5	4.9	4.9	4.2	3.7	4.0	3.9	3.4	3.3
Precipitation	PrecAvg	25.1	1.5	1.1	1.2	1.3	1.7	3.1	3.9	3.4	2.4	2.1	1.8	1.5
	PrecMax	34.3	2.8	1.9	2.1	2.4	3.7	5.5	8.2	5.5	5.4	5.7	4.4	3.3
	PrecMin	19.8	0.2	0.1	0.2	0.3	0.6	0.7	0.9	1.8	0.6	0.4	0.6	0.4
	PrecStd	3.9	0.7	0.5	0.5	0.7	0.8	1.3	2.0	1.1	1.3	1.3	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	42.7	48.0	61.0	75.1	78.1	81.0	76.6	67.2	57.4	48.7	43.2
		MCWB	37.2	37.4	39.5	46.4	55.5	57.8	63.1	60.9	55.3	50.5	45.4	39.3
	2%	DB	38.9	40.2	44.0	55.8	69.6	74.6	76.9	72.4	63.2	52.6	43.2	40.4
		MCWB	35.5	35.6	38.4	44.2	53.8	56.3	61.7	59.6	54.2	47.3	41.2	37.0
	5%	DB	35.8	37.8	41.0	51.2	64.1	69.7	73.2	68.4	59.3	49.4	39.8	38.1
		MCWB	32.9	34.1	35.8	41.9	50.8	55.3	60.0	57.9	52.7	45.0	37.6	35.3
	10%	DB	32.6	34.3	38.3	46.9	58.3	64.8	69.3	64.6	56.1	46.6	37.0	34.8
		MCWB	30.5	31.2	33.9	39.4	47.8	53.0	58.2	56.5	50.5	43.6	35.1	32.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	38.1	40.9	48.7	56.6	61.5	65.3	63.7	58.1	52.2	45.7	40.0
		MCDB	41.0	41.8	45.7	59.3	69.5	71.2	74.9	70.5	63.7	56.0	48.5	42.5
	2%	WB	35.7	36.1	38.7	44.8	54.0	58.5	62.9	61.4	55.4	48.1	41.2	37.3
		MCDB	38.7	39.6	44.3	54.1	64.1	68.6	73.1	68.4	61.0	51.3	43.0	39.9
	5%	WB	33.3	34.2	36.3	42.5	51.4	56.4	61.0	59.5	53.3	45.8	37.8	35.3
		MCDB	35.5	37.8	40.8	49.8	60.3	67.1	70.2	66.2	58.5	48.8	39.3	37.8
	10%	WB	30.6	31.7	34.3	40.1	48.3	54.4	59.1	57.6	51.2	43.6	35.2	32.5
		MCDB	32.5	34.5	38.1	46.4	55.9	63.9	66.3	63.6	55.4	46.5	36.7	34.4
Mean Daily Temperature Range	5% DB	MDBR	16.8	18.9	20.3	19.9	21.8	21.5	21.4	21.2	18.8	15.9	13.9	15.9
		MCDBR	15.4	15.4	19.4	28.8	34.1	32.5	30.0	29.3	25.7	21.5	15.0	16.0
		MCWBR	13.5	12.4	14.1	17.8	18.4	16.7	16.1	17.6	17.5	16.1	13.0	14.2
	5% WB	MCDBR	15.4	14.9	18.0	25.3	27.9	27.2	26.5	24.7	22.5	19.5	14.3	16.5
		MCWBR	13.7	12.2	13.3	16.1	16.8	15.0	15.2	15.8	16.0	15.4	12.8	14.7
Clear-Sky Solar Irradiance	taub	0.200	0.258	0.287	0.320	0.329	0.325	0.340	0.329	0.299	0.284	0.213	N/A	N/A
	taud	2.077	2.271	2.318	2.394	2.477	2.518	2.489	2.507	2.574	2.489	2.151	N/A	N/A
	Ebn at Noon	154	226	262	276	282	285	277	271	260	218	145	N/A	N/A
	Edh at Noon	10	18	26	30	30	29	30	27	20	15	9	N/A	N/A
All-Sky Solar Radiation	RadAvg	66	254	709	1271	1489	1625	1501	1149	714	311	91	33	
	RadStd	6	37	66	103	122	135	177	104	77	44	15	3	